



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES
CHENNAI - 602105



AI InterviewX – an AI-powered mock interview platform

(Agile Sprint Planning & User Story Workshop)

CSA1012-SOFTWARE ENGINEERING

BACHELOR OF ENGINEERING

In

Computer Science Engineering

Submitted by

Y. Praveen Raj (192372306)

Under the Supervision of

Dr. K. Anita Davamani

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1. Project Vision and Stakeholders

Project vision

AI InterviewX is an AI-powered mock interview platform that helps job seekers, especially final-year engineering students and early-career professionals, practice technical and behavioural interviews on demand. The platform simulates a real interview using AI-generated questions tailored to a chosen role and difficulty level, evaluates the candidate's spoken or written responses, and returns structured feedback so the candidate can identify and close skill gaps before a real interview.

Objectives:

- Give candidates unlimited, low-cost practice interviews tailored to a specific job role.
- Provide objective, structured feedback across communication, technical accuracy, and confidence.
- Let candidates track improvement across multiple sessions over time.
- Keep the question bank current and relevant for admins/content owners to maintain.

Key stakeholders

Stakeholder	Interest / role
Job candidates (end users)	Primary users; need realistic practice and actionable feedback to prepare for real interviews.
Platform admin / content team	Maintains and updates the question bank per role and difficulty.
Product owner	Owns the backlog, prioritises features based on user value.
Development team	Builds and delivers the sprint backlog; estimates and executes tasks.
Career mentors / placement cell	Indirect stakeholders who may review candidate reports as a coaching aid.

2. Epics and User Stories

Epics

- Epic 1: Interview setup – letting a candidate configure and start a relevant mock interview.
- Epic 2: Interview delivery – conducting the interview through text/voice question-and-answer.
- Epic 3: Feedback and scoring – evaluating answers and presenting structured feedback.
- Epic 4: Progress tracking – letting candidates review history and export reports.
- Epic 5: Content administration – letting admins manage the question bank.
- Epic 6: Account and security – authentication and data privacy.

User stories

ID	User story	Priority	Story points
US-1	As a job candidate, I want to select a target role and difficulty level, so that the mock interview matches the job I am preparing for.	High	3
US-2	As a job candidate, I want the AI to ask me interview questions one at a time through text or voice, so that the session feels like a real interview.	High	5
US-3	As a job candidate, I want to receive feedback on my answers immediately after the interview, so that I can identify weak areas.	High	8
US-4	As a job candidate, I want to see a score breakdown across communication, technical accuracy, and confidence, so that I understand where to improve.	Medium	5
US-5	As a returning user, I want to view my past interview sessions and score history, so that I can track my improvement over time.	Medium	3
US-6	As a job candidate, I want to download a PDF report of my interview performance, so that I can review it offline or share it with a mentor.	Low	2
US-7	As a platform admin, I want to add and edit the question bank for each role, so that the interview content stays current.	Medium	5
US-8	As a job candidate, I want to sign up and log in securely, so that my interview history is private and saved to my account.	High	3

3. Acceptance Criteria

Acceptance criteria are written in Given–When–Then format for the highest-priority user stories selected for Sprint 1.

US-1: Select target role and difficulty

- Given I am on the home screen, When I select a role (e.g. "AI Engineer") and a difficulty (Beginner/Intermediate/Advanced), Then the system loads a question set matching that role and difficulty.
- Given I have not selected a role, When I try to start the interview, Then the system shows a validation message and blocks the start.

US-2: Conduct the mock interview

- Given a question set has loaded, When the session starts, Then the AI presents one question at a time and waits for my response before showing the next.
- Given I am answering by voice, When I finish speaking, Then the system transcribes my answer and displays it before moving on.

US-3: Receive post-interview feedback

- Given I have completed all questions in the session, When the interview ends, Then the system generates feedback for each answer within 30 seconds.
- Given feedback has been generated, When I view it, Then each answer shows at least one strength and one improvement area.

US-4: View score breakdown

- Given feedback has been generated, When I open the results screen, Then I see separate scores for communication, technical accuracy, and confidence, each out of 10.

US-8: Secure sign up and login

- Given I am a new user, When I sign up with email and password, Then my account is created and I am logged in automatically.
- Given I enter an incorrect password, When I try to log in, Then the system rejects the attempt and shows an error without revealing which field was wrong.

4. Product Backlog Prioritization

The backlog is prioritised using a value-versus-effort view: stories that are essential to a working end-to-end interview flow (setup, delivery, feedback, authentication) are ranked High; stories that improve the experience but are not required for a first usable release (score breakdown detail, history) are ranked Medium; and nice-to-have additions (PDF export) are ranked Low.

Priority	Rationale	Stories
High	Required for a candidate to complete one full mock interview end to end.	US-1, US-2, US-3, US-8
Medium	Improves depth of feedback and retention but not required for the first working release.	US-4, US-5, US-7
Low	Adds convenience; deferrable to a later release.	US-6

Dependency note: US-2 (interview delivery) depends on US-1 (setup) to know which question set to load. US-3 (feedback) depends on US-2 producing recorded answers. US-8 (authentication) has no functional dependency but is required first so that sessions can be saved against a user account.

5. Sprint Backlog (Sprint 1)

Sprint 1 selects the High-priority stories needed to deliver one complete, working interview flow: account creation, role/difficulty selection, the interview itself, and feedback.

US-1 – task breakdown

- Design role/difficulty selection UI
- Build question-bank filter API
- Wire selection screen to API

US-8 – task breakdown

- Set up auth service (signup/login)
- Add password hashing and session tokens
- Build login/signup screens

US-2 – task breakdown

- Build question delivery flow (text)
- Integrate speech-to-text for voice answers
- Build answer capture and session state

US-3 – task breakdown

- Integrate LLM for answer evaluation
- Design feedback generation prompt/logic
- Build feedback display screen

6. Story Point Estimation

Estimation uses the Fibonacci sequence (1, 2, 3, 5, 8, 13) via Planning Poker, where the team scores each story independently and discusses outliers before converging on a value. Points reflect relative complexity and uncertainty, not hours.

Story	Justification	Points
US-1	Simple UI plus a filtered query against the question bank; low complexity, few unknowns.	3
US-8	Standard auth flow, but security handling (hashing, sessions, validation) adds moderate complexity.	3
US-2	Requires session-state management plus speech-to-text integration – higher complexity and external dependency.	5
US-3	Depends on LLM integration for evaluation quality, which carries the most uncertainty in the sprint.	8

Sprint 1 total: 19 story points across 4 stories.

7. Sprint Goal and Expected Deliverables

Sprint goal

By the end of Sprint 1, a candidate can create an account, select a target role and difficulty, complete a full mock interview, and receive AI-generated feedback on their answers – the minimum end-to-end flow that makes the product usable.

Expected deliverables

- Working signup/login flow with secure session handling (US-8).
- Role and difficulty selection screen connected to a filtered question bank (US-1).
- Functional interview delivery flow supporting text and voice answers (US-2).
- AI-generated feedback shown to the candidate at the end of the session (US-3).
- Sprint demo showing one complete interview session, start to finish.

Out of scope for Sprint 1

- Detailed score breakdown by category (US-4) – planned for Sprint 2.
- Session history and progress tracking (US-5) – planned for Sprint 2.
- PDF report export (US-6) – planned for a later release.
- Admin question-bank management UI (US-7) – planned for Sprint 2.

References

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